

EDUCATION	School of Computer Science, University of Sydney	Sydney, Australia
	<i>Ph.D., Computer Science</i>	2023 - 2026
	• Advisors: Chang Xu & Daochang Liu • Research area: LLM Uncertainty Quantification and Confidence Calibration • Expected thesis submission: 31 December 2026	
	School of Computer Science, University of Sydney	Sydney, Australia
	<i>M.Phil., Computer Science</i>	2021 - 2023
	• Advisors: Chang Xu & Lijun Chang • Research area: Uncertainty Quantification, Confidence Calibration, Neural Architecture Search	
	School of Computer Science, University of Sydney	Sydney, Australia
	<i>M.E., Data Science</i>	2020 - 2021
	Huazhong University of Science and Technology	Wuhan, China
	<i>B.E., Communication Engineering</i>	2013 - 2017
	• Received Science and Technology Scholarship	
EXPERIENCES	Research Intern Apple MLR, Cambridge UK	2026.05 - 2026.10
	• Multi-turn RL post-training of LLM agents for information gathering and tool use. • Reward design, environment and tool-calling improvements, with rollout audits for reward hacking. • LLM judge as the reward source: accuracy, prompting, and parsing robustness. • Training efficiency: async versus sync rollouts, sample utilisation, and OOM fixes.	
	Research Intern Google Research, Sydney AU	2025.11 - 2026.02
	• Speculative decoding for efficient LLM inference.	
	Teaching Assistant University of Sydney, Sydney	2022 - now
	• COMP5329 Deep Learning, COMP5318 Machine Learning and Data Mining, OCMP5329 Deep Learning (Online), HTIN5005 Applied Healthcare Data Science, BUSS6002 Data Science in Business, QBUS5010 Dashboarding and Data Visualisation.	
	Construction Supervisor Songjiang Construction Union, Shanghai	2017.07 - 2019.10
	IOS Developer Intern Trip.com, Shanghai	2015.06 - 2015.09
	Frontend Developer Intern Sunallies, Shanghai	2014.06 - 2014.09
SELECTED PUBLICATIONS	1. Linwei Tao, Yi-Fan Yeh, Mingjing Dong, Tao Huang, Philip Torr, Chang Xu. Revisiting Uncertainty Estimation and Calibration of Large Language Models. <i>NeurIPS 2025 Scaling Environments for Agents (SEA) Workshop</i>. 2. Linwei Tao, Yi-Fan Yeh, Bo Kai, Mingjing Dong, Tao Huang, Tom A. Lamb, Jialin Yu, Philip H.S. Torr, Chang Xu. Can Large Language Models Express Uncertainty Like Human? <i>Preprint</i>, 2026. 3. Younan Zhu, Linwei Tao, Mingjing Dong, Chang Xu. Attention Calibration for Reducing Hallucination in Large Vision-Language Models. <i>Preprint</i>, 2026. 4. Linwei Tao, Younan Zhu, Haolan Guo, Mingjing Dong, Chang Xu. A Benchmark Study on Calibration. <i>International Conference on Learning Representations, ICLR 2024</i>.	

AWARDS AND GRANTS	• Apple Scholars in AI/ML PhD Fellowship (20 selected worldwide, first Australian recipient), Apple 2026.04
	• FFT Student Survey Award for Outstanding Achievement in BUSS6002 , USYD Business School 2025.12
	• AAAI 2026 Student Program Committee (3 selected worldwide), AAAI 2025.06
	• International Tuition Fee Scholarship , USYD 2023.06
	• Faculty of Engineering Research Support Scholarship , USYD 2023.05
	• Excellent Student Cadre , HUST, 2015.09
	• Science and Technology Scholarship , HUST 2014.06
OTHER PUBLICATIONS	1. Haolan Guo, Linwei Tao , Haoyang Luo, Minjing Dong, Chang Xu. Sample Margin-Aware Recalibration of Temperature Scaling. <i>International Conference on Machine Learning, ICML</i> 2026.
	2. Xiaoyang Li, Linwei Tao , Haohui Lu, Minjing Dong, Junbin Gao, Chang Xu. Calibrating Graph Neural Networks with Wavelet-Aware Temperature Scaling. <i>International Conference on Learning Representations, ICLR</i> 2026.
	3. Lei Pan, Zhipeng Lu, Yuan Zheng, Chongyao Yan, Haitao Wen, Linwei Tao , Chang Xu. Task-adaptive continual learning of vision language models via prototype routing and prompt. <i>Neurocomputing</i> , 2026.
	4. Yingqing Yuan, Linwei Tao , Haohui Lu, Matloob Khushi, Imran Razzak, Mark Dras, Jian Yang, Usman Naseem. KG-UQ: Knowledge Graph-Based Uncertainty Quantification for Long Text in Large Language Models. <i>International World Wide Web Conference, WWW 2025 Workshop on Sustainable AI for the Future Web</i>
	5. Haoyang Luo, Linwei Tao , Minjing Dong, Chang Xu. Beyond One-Hot Labels: Semantic Mixing for Model Calibration. <i>International Conference on Machine Learning, ICML</i> 2025.
	6. Jinxu Lin, Linwei Tao , Minjing Dong, Chang Xu. Uncertainty Weighted Gradients for Model Calibration. <i>The IEEE / CVF Computer Vision and Pattern Recognition Conference, CVPR</i> 2025.
	7. Jinxu Lin, Linwei Tao , Minjing Dong, Chang Xu. Diffusion Attribution Score: Evaluating Training Data Influence in Diffusion Model. <i>International Conference on Learning Representations, ICLR</i> 2025. (Spotlight, 5.1% of 11,500 submissions)
	8. Linwei Tao , Minjing Dong, Chang Xu. Feature Clipping for Uncertainty Calibration. <i>AAAI Conference on Artificial Intelligence, AAAI</i> 2025.
	9. Lei Pan, Wuyang Luan, Yuan Zheng, Junhui Li, Linwei Tao , Chang Xu. GraphFusion: Integrating Multi-Level Semantic Information with Graph Computing for Enhanced 3D Instance Segmentation. <i>Neurocomputing</i> , 2024.
	10. Linwei Tao , Minjing Dong, Chang Xu. Dual Focal Loss for Calibration. <i>International Conference on Machine Learning, ICML</i> 2023.
	11. Linwei Tao , Minjing Dong, Daochang Liu, Changming Sun, Chang Xu. Calibrating a Deep Neural Network with Its Predecessors. <i>International Joint Conference on Artificial Intelligence, IJCAI</i> 2023.

PREPRINTS

1. Yunke Wang, **Linwei Tao**, Bo Du, Yutian Lin, Chang Xu. Visual Imitation Learning with Calibrated Contrastive Representation.
2. **Linwei Tao**, Haoyang Luo, Minjing Dong, Chang Xu. Confidence Calibration under Ambiguous Ground Truth. *arXiv:2603.22879*, 2026.
3. **Linwei Tao**, Haolan Guo, Minjing Dong, Chang Xu. Consistency Calibration: Improving Uncertainty Calibration via Consistency among Perturbed Neighbors.
4. Rui Xu, Yunke Wang, **Linwei Tao**, Wenjie Xuan, Yong Luo. Diversifying Personalized Research Ideation against AI-Induced Homogenization. *arXiv:2607.28087*, 2026.

ACADEMIC
SERVICES

Conference Reviewer: *ICML, NeuIPS, ICLR, CVPR, ICCV, AAAI, IJCAI, ACMMM, AISTATS*

Journal Reviewer: *T-MM, TMLR, DAMI, TPAMI, Machine Learning*